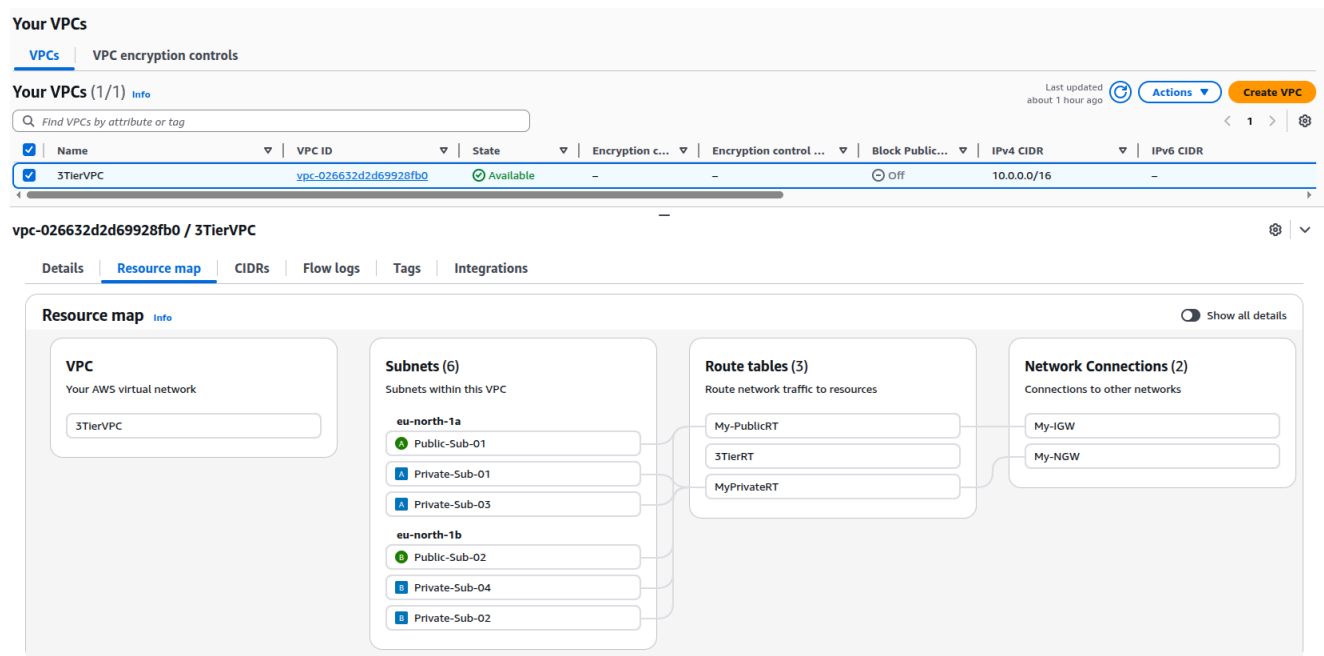


Objective : Deploy a 3-tier web application using AWS with a Public ALB in the presentation layer, ASG with 2 EC2 instances in the application layer, and an RDS instance in the database layer.

1. Create VPC & Subnets: Set up public & private subnets, route tables, and internet gateway.

Ans :



I created my VPC with two Public Subnets in different AZs and two Private Subnets in different AZs. I created my internet gateway and attached to the public subnets and then created NAT gateway in public subnet and then attached it to the private subnets.

2. Launch RDS: Deploy an RDS instance in the private subnet and configure access from EC2 instances.

Ans: I launched my RDS using Multi AZ instance deployment and deployed in the private subnets created by me . I created an temporary ec2 instance then through it connected the mysql client in RDS instance which i created to create the table .

crud-db

Summary

DB identifier

crud-db

CPU

49.91%

Status

Configuring-enhanced-monitoring

Class

db.t3.micro

Role

Instance

Current activity

Engine

MySQL Community

Region & AZ

eu-north-1b

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Tags

Recommendations

Connect using

Code snippets

Use when connecting through SDK, APIs, or third-party tools including agents.

CloudShell

Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

Endpoints

Use when connecting through any IDE interface.

Internet access gateway

Disabled

IAM Authentication

Disabled

Programming language

MySQL (Linux)

Endpoint type

Instance endpoint

Connection steps

Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

1 mysql -h crud-db.czggwgqayuae.eu-north-1.rds.amazonaws.com -P 3306 -u admin -p'<Enter_DB_Password>' --ssl-verify-server-cert --ssl-ca=/certs/global-bundle.pem mysql

Database changed

mysql> show tables;

Tables_in_hello_world |

messages |

1 row in set (0.00 sec)

mysql> select * from messages;

id | message | created_at |

1 | Hello from the database! | 2026-02-05 16:45:43 |

2 | Welcome to our three-tier architecture demo | 2026-02-05 16:45:43 |

3 | This is a simple example showing frontend, backend, and database | 2026-02-05 16:45:43 |

4 | Hello , Testing! | 2026-02-05 17:54:50 |

4 rows in set (0.00 sec)

mysql>

3. Set Up EC2 Instances: Create a launch template and configure ASG with 2 EC2 instances running the app in private subnets.

Ans : I created launch templates separately for my frontend and backend layers and configured the instances. The instances and the application will run in the private subnets. The frontend template is :

frontend-new (lt-052b0abbb7eeacbbc)

Actions

Delete template

Launch template details

Launch template ID

lt-052b0abbb7eeacbbc

Launch template name

frontend-new

Default version

1

Owner

am:aws:iam::596059883112:root

Details

Versions

Template tags

Launch template version details

Version

1 (Default)

Description

New Launch Template

Date created

2026-02-05T17:51:06.000Z

Created by

am:aws:iam::596059883112:root

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID

ami-04233b5aeece09244

Instance type

t3.small

Availability Zone

-

Availability Zone Id

-

Key pair name

Plyush

Security groups

-

Security group IDs

sg-08a48ee48f6333df1

The backend template is :

backend (lt-0d25b7ea21f96356d)

ActionsDelete template

Launch template details

Launch template ID

lt-0d25b7ea21f96356d

Launch template name

backend

Default version

1

Owner

arn:aws:iam::596059883112:root

DetailsVersionsTemplate tags

Launch template version details

ActionsDelete template version

Version

1 (Default)

Description

Launch template for backend

Date created

2026-02-05T17:06:41.000Z

Created by

arn:aws:iam::596059883112:root

Instance detailsStorageResource tagsNetwork interfacesAdvanced details

AMI ID

arn:aws:ec2:eu-north-1:596059883112:ami-04233b5aece09244

Instance type

t3.small

Availability Zone

-

Availability Zone Id

-

Key pair name

Plyush

Security groups

-

Security group IDs

sg-0a644300e0dcada7b

These are configured in such manner that they will create the ec2 instances in different AZs and in different private subnets.

Now the ASG for frontend is :

frontend-new-asg

Edit

frontend-new-asg Capacity overview

arn:aws:autoscaling:eu-north-1:596059883112:autoScalingGroup:c5cce54b-6106-4505-bd5d-9b960d6a9878:autoScalingGroupName/frontend-new-asg

Desired capacity

2

Scaling limits

2 - 3

Desired capacity type

Units (number of instances)

Status

-

Date created

Thu Feb 05 2026 23:22:24 GMT+0530 (India Standard Time)

DetailsIntegrationsAutomatic scalingInstance managementInstance refreshActivityMonitoringTags - moved

Launch template

Edit

Launch template

lt-052b0aabb7eeacbbc
frontend-new

AMI ID

arn:aws:ec2:eu-north-1:596059883112:ami-04233b5aece09244

Instance type

t3.small

Owner

arn:aws:iam::596059883112:root

Version

Default

Security groups

-

Security group IDs

sg-08a48ee48f6333df1

Create time

Thu Feb 05 2026 23:21:06 GMT+0530 (India Standard Time)

Description

New Launch Template

Storage (volumes)

-

Key pair name

Plyush

Request Spot Instances

No

View details in the launch template console

Network

Edit

Availability Zones

eun1-az1 (eu-north-1a)
eun1-az2 (eu-north-1b)
eun1-az1 (eu-north-1a)

Subnet ID

subnet-07d1352941da01430
subnet-0a20f28624959cc32

Availability Zone distribution

Balanced best effort

The ASG for Backend is :

backend-asg

Edit

backend-asg Capacity overview

arn:aws:autoscaling:eu-north-1:596059883112:autoScalingGroup:561f22f3-5a1b-4a31-9e0c-721c801cba4d:autoScalingGroupName/backend-asg

Desired capacity

2

Scaling limits

2 - 3

Desired capacity type

Units (number of instances)

Status

-

Date created

Thu Feb 05 2026 22:39:18 GMT+0530 (India Standard Time)

DetailsIntegrationsAutomatic scalingInstance managementInstance refreshActivityMonitoringTags - moved

Launch template

Edit

Launch template

lt-0d25b7ea21f96356d
backend

AMI ID

arn:aws:ec2:eu-north-1:596059883112:ami-04233b5aece09244

Instance type

t3.small

Owner

arn:aws:iam::596059883112:root

Version

Default

Security groups

-

Security group IDs

sg-0a644300e0dcada7b

Create time

Thu Feb 05 2026 22:36:41 GMT+0530 (India Standard Time)

Description

Launch template for backend

Storage (volumes)

-

Key pair name

Plyush

Request Spot Instances

No

View details in the launch template console

Network

Edit

Availability Zones

eun1-az1 (eu-north-1a)
eun1-az2 (eu-north-1b)

Subnet ID

subnet-07d1352941da01430

Availability Zone distribution

Balanced best effort

4. Configure ALB: Set up an Application Load Balancer in the public subnet, link to EC2 instances via target group, and configure security groups.

Ans:

I created two application load balancers one for frontend which handles the internet requests and forward to the ec2 instances , it is configured as internet facing. The backend load balancer is internal facing which handles the backend reuquests to the ec2 instances in the private subnet.

The image displays two screenshots of the AWS Management Console, specifically the 'Details' and 'Listeners and rules' sections for two Application Load Balancers (ALBs).

frontend-alb Details:

- Load balancer type:** Application
- Scheme:** Internet-facing
- Status:** Active
- Hosted zone:** Z23TAZ6LKFMNIO
- VPC:** vpc-026632d2d69928fb0
- Availability Zones:** eu-north-1b (eun1-az2), eu-north-1a (eun1-az1)
- Load balancer IP address type:** IPv4
- Date created:** February 5, 2026, 22:19 (UTC+05:30)
- Load balancer ARN:** arn:aws:elasticloadbalancing:eu-north-1:596059883112:loadbalancer/app/frontend-alb/66f72c1fb85ab0c2
- DNS name:** frontend-alb-1881685455.eu-north-1.elb.amazonaws.com (A Record)

Listeners and rules (1):

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
HTTP:80	Forward to target group frontend (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applicable

backend-alb Details:

- Load balancer type:** Application
- Scheme:** Internal
- Status:** Active
- Hosted zone:** Z23TAZ6LKFMNIO
- VPC:** vpc-026632d2d69928fb0
- Availability Zones:** eu-north-1b (eun1-az2), eu-north-1a (eun1-az1)
- Load balancer IP address type:** IPv4
- Date created:** February 5, 2026, 22:20 (UTC+05:30)
- Load balancer ARN:** arn:aws:elasticloadbalancing:eu-north-1:596059883112:loadbalancer/app/backend-alb/72581846dc5797ff
- DNS name:** internal-backend-alb-140462122.eu-north-1.elb.amazonaws.com (A Record)

Listeners and rules (1):

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
HTTP:80	Forward to target group backend (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applicable

The target groups associated with frontend and backend load balancers are :

Target group: backend

Details

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (1)

Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 >

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Administrative ov...	Override details	Launch...
☐	I-Oda20ad05ca4036f7	-	80	eu-north-1a (e...	Healthy	-	No override	No override is curren...	February .

Target group: frontend

Details

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 >

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Administrative ov...	Override details	Launch...
☐	I-Od65fa0844859338f	-	80	eu-north-1b (e...	Healthy	-	No override	No override is curren...	February .
☐	I-Oa27dccd020423728	-	80	eu-north-1a (e...	Healthy	-	No override	No override is curren...	February .

5. Security & IAM: Set up appropriate security groups for each layer and IAM roles for EC2 instances.

Ans :

The Security groups associated with each load balancers and asgs are as follows :

Frontend ALB -SG

sg-Off4fe6f31e26e385 - frontend-alb-sg

Actions

Details

Security group name

frontend-alb-sg

Security group ID

sg-off4fe6f31e26e385

Description

Allow

VPC ID

vpc-026632d2d69928fb0

Owner

596059883112

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

☐

Name

☐

-

☐

Security group rule ID

☐

sgr-off9302d36846af57

☐

IP version

☐

IPv4

☐

Type

☐

HTTP

☐

Protocol

☐

TCP

☐

Port range

☐

80

☐

Source

☐

0.0.0.0/0

☐

Description

☐

-

sg-Off4fe6f31e26e385 - frontend-alb-sg

Actions

Details

Security group name

frontend-alb-sg

Security group ID

sg-Off4fe6f31e26e385

Description

Allow

VPC ID

vpc-026632d2d69928fb0

Owner

596059883112

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Outbound rules (1)

Manage tags

Edit outbound rules

Search

☐

Name

☐

-

☐

Security group rule ID

☐

sgr-0323a1204baa001c3

☐

IP version

☐

-

☐

Type

☐

All traffic

☐

Protocol

☐

All

☐

Port range

☐

All

☐

Destination

☐

sg-08a48ee48f6333df1...

☐

Description

☐

-

Frontend ASG-SG

Actions ▼

VPC ID
[ypc-026632d2d69928fb0](#)

Inbound rules

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Actions ▼

VPC ID
[vpc-026632d2d69928fb0](#) 

Inbound rules

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Backend ALB-SG

Actions ▼

VPC ID
[vpc-026632d2d69928fb0](#)

Inbound rules

sgl-00453110750714050	All drink	All	All	sgl-00453110750714050...
-----------------------	-----------	-----	-----	--------------------------

Actions ▼

VPC ID
[vpc-026632d2d69928fb0](#)

Inbound rules

sgf-014a18835203b4c9b	All traffic	All	All	sg-0a8443008b0c0ada7...
-----------------------	-------------	-----	-----	---

Backend ASG-SG

Actions ▼

VPC ID
[vpc-026632d2d69928fb0](#)

Inbound rules

sgl-0454c6645502e35e3	HTTP	TCP	80	sg-01055557370e60e03...
-----------------------	------	-----	----	-------------------------

sg-0a644300e0dcada7b - backend-asg-sg

Actions

Details

Security group name

backend-asg-sg

Security group ID

sg-0a644300e0dcada7b

Description

allow

VPC ID

vpc-026632d2d69928fb0

Owner

596059883112

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Outbound rules (1)

Search

Manage tags

Edit outbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Destination	Description
	-	sgr-00d18a5753bb9b52d	-	MySQL/Aurora	TCP	3306	sg-0376ad0931e59a2b6	-

RDS SG

sg-0376ad0931e59a2b6 - db-sg

Actions

Details

Security group name

db-sg

Security group ID

sg-0376ad0931e59a2b6

Description

allow

VPC ID

vpc-026632d2d69928fb0

Owner

596059883112

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Search

Manage tags

Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
	-	sgr-0af704f96786c8009	-	MySQL/Aurora	TCP	3306	sg-0a644300e0dcada7b	-

6.Test & Validate: Ensure EC2 instances are healthy, ALB routes traffic, and auto-scaling works as expected.

Ans:

Checking the 2 EC2 instances health :

My EC2 instances created by the frontend ASG is healthy and working fine .

Target group: frontend

Details

Targets

Monitoring

Health checks

Attributes

Tags

Details

arn:aws:elasticloadbalancing:eu-north-1:596059883112:targetgroup/frontend/5ec28a198390e26e

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-026632d2d69928fb0

IP address type

IPv4

Load balancer

frontend-alb

2

Total targets

2

Healthy

0

Unhealthy

0

Unused

0

Initial

0

Draining

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Same goes with my backend as well :

Target group: backend

Details

Targets

Monitoring

Health checks

Attributes

Tags

Details

arn:aws:elasticloadbalancing:eu-north-1:596059883112:targetgroup/backend/b73c1a1b20426cc1

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-026632d2d69928fb0

IP address type

IPv4

Load balancer

backend-alb

1

Total targets

1

Healthy

0

Unhealthy

0

Unused

0

Initial

0

Draining

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

The autoscaling groups works fine as well for the frontend and backend.

Auto Scaling group: frontend-new-asg

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags - moved

Instances (2)

Filter Instances

< 1 >

☐	Instance ID	Lifecycle	Instance type	Weighted capacity	Launch template...	Availability Zone	Health status	Protected from
☐	i-0178a0310220b4d61	InService	t3.small	-	frontend-new Version 1	eun1-az2 (eu-north-...	Healthy	
☐	i-01b9edf622eb9feff	InService	t3.small	-	frontend-new Version 1	eun1-az1 (eu-north-...	Healthy	

Instance lifecycle policy for lifecycle hooks

Controls instance behavior when an instance transitions through its lifecycle states.

Termination hook abandon behavior

Terminate (default)

Manage policy

Lifecycle hooks (0)

Actions

Create lifecycle hook

Auto Scaling group: backend-asg

Instances (2)

Filter Instances

< 1 >

☐	Instance ID	Lifecycle	Instance type	Weighted capacity	Launch template...	Availability Zone	Health status	Protected from
☐	i-0538f4ae8850dc655	InService	t3.small	-	backend Version 1	eun1-az2 (eu-north-...	Healthy	
☐	i-0da20ad05ca4036f7	InService	t3.small	-	backend Version 1	eun1-az1 (eu-north-...	Healthy	

Instance lifecycle policy for lifecycle hooks

Controls instance behavior when an instance transitions through its lifecycle states.

Termination hook abandon behavior

Terminate (default)

Manage policy

Lifecycle hooks (0)

Filter lifecycle hooks

Actions

Create lifecycle hook

The final 3 Tier Application after deployment looks like this :

CloudKeeper

Basic Three-Tier Architecture Example

Message from Database:

- Hello , Testing! (2026-02-05 17:54:50)
- Hello from the database! (2026-02-05 16:45:43)
- Welcome to our three-tier architecture demo (2026-02-05 16:45:43)
- This is a simple example showing frontend, backend, and database (2026-02-05 16:45:43)

Add New Message

New Message:

Save Message